stationed around the L1 point between the earth and the sun, which is nearly 1.5 million km away from the earth. The aim of the mission is to demonstrate the technology needed to establish a space-based observatory to study gravitational waves. Therefore, the correct answer is (d).

79. What is the importance of using Pneumococcal Conjugate Vaccines in India?

1. These vaccines are effective against pneumonia as well as meningitis and sepsis.
2. Dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.
3. These vaccines have no side effects and cause no allergic reactions.

Select the correct answer using the code given below:

(a) 1 only (b) 1 and 2 only
(c) 3 only (d) 1, 2 and 3

Ans. (b) : Pneumococcus is a leading cause of bacterial pneumonia, meningitis and sepsis in children. The burden of pneumococcal meningitis is such that it constitutes about half of all childhood meningitis cases in most settings and a greater proportion of meningitis deaths. The vaccine will protect children against severe forms of pneumococcal diseases, such as pneumonia, meningitis and sepsis. The vaccine also prevents antibiotic-resistant pneumococcal infections caused by vaccine serotypes. Hence statements 1 and 2 are correct. However, anyone who has ever had a life threatening allergic reaction to a dose of this vaccine, to an earlier pneumococcal vaccine should not get it. Anyone with a severe allergy to any component of it should not get the vaccine. With any medicine including vaccines, there is a chance of side effects. These are usually mild and go away on their own, but serious reactions are also possible. Hence, statement 3 is incorrect.

80. In India, the term "Public Key Infrastructure" is used in the context of-

(a) Digital security infrastructure
(b) Food security infrastructure
(c) Health care and education infrastructure
(d) Telecommunication and transportation infrastructure

Ans. (a): Public key infrastructure (PKI) is a technology for authenticating users and devices in the digital world. The basic idea is to have one or more trusted parties digitally sign documents certifying that a particular cryptographic key belongs to a particular user or device. Hence option (a) is correct.

81. In the correct of recent advances in human reproductive technology, "Pronuclear Transfer" is used for

(a) fertilization of egg in vitro by the donor sperm
(b) genetic modification of sperm producing cells
(c) development of stem cells into functional embryos
(d) prevention of mitochondrial diseases in offspring

Ans. (d) : The mother's egg is first fertilized with the father's sperm, producing a zygote in pronuclear transfer. The pronuclei of the egg and sperm are then removed from the zygote and inserted into a donor egg that has been fertilized and has had its own nucleus removed (a pronucleus is the nucleus of the egg or sperm at the stage of fertilization prior to nucleus fusion). The zygote derived from the donor egg is then implanted into the mother's uterus. It is not fully in vitro.
Option (d) is correct. This is a new strategy to prevent the transmission of mitochondrial diseases in offspring.

82. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

(a) 1, 2, 3 and 5 only (b) 1, 3 and 4 only
(c) 2, 4 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (b) : Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. Artificial intelligence has various applications in today's society. In multiple industries, such as healthcare, entertainment, finance, education etc. AI has been used in the text-to-speech conversion. Hence statement 1, 3, and 4 are correct.

83. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm
2. VLC is known as long-range optical wireless communication
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

Ans. (c) : Visible Light Communication (VLC) systems employ visible light for communication that occupies the spectrum from 380 nm to 750 nm. The technology uses fluorescent lamps (ordinary lamps, not special communications devices) to transmit signals at 10 kbit/s or LEDs for upto 500 Mbit/s over short distances. Visible light communication is an emerging technology, which provides distinct facilities such as fast data communication, secure data communication; high data rate wireless communication, instead of radio frequency, VLC uses light to transfer data. The visible light spectrum is unlicensed and 10,000 times larger than the range of radio frequencies. It can be used to alternate to the existing ratio based wireless communication technologies or in hybrid. Moreover, integrated VLC resolve the issue of load balancing in existing wireless networks. Light fidelity is a recent technology under VLC, which can be used for wireless communication as it has many advantages. Such as no electromagnetic interference, low cost and high data transfer rate, much higher than traditional bluetooth. Hence, the correct option is (c).

84. With reference to "Blockchain Technology", consider the following statements:

1. It is a public ledger that everyone can inspect, but with single user controls
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only
3. Applications that depend on basic features of blockchain can be developed without anybody's permission

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 2 only
(c) 2 only (d) 1 and 3 only

Ans. (d) : Blockchain can be defined as a distributed ledger technology that can record transactions between parties securely and permanently. By 'sharing' databases between multiple parties, blockchain essentially removes the need for intermediaries who were previously required to act as trusted third parties to verify records and coordinate transactions. It is a foundational technology or a platform that allows designing a secure way to record transactions and circulate it among signatories, or any kind of target group with an internet connection. At its core, it is an extremely democratic ledger that cannot be arbitrarily manipulated and is easily shareable. Hence, statement 1 is correct.

The structure of blockchain to eliminate intermediary and enhance authentication such that it supports a wide range of applications in health, education, governance, diamond industry, financial sector etc. Hence, statement 2 is not correct.
Most of the applications using blockchain technology do not fulfill the criteria of patent filing in blockchain technology. Distributed ledger in peer to peer network is used for recording the transaction which does not generate much confidence in getting the patent as it is like getting a patent for an algorithm/computer program and as per Indian Patent Act, 1970 Section 3(k) which states that "a mathematical or business method or a computer program per se or algorithms are not patentable". As per the ruling of Delhi High Court in the 2014 case of *Ericson Vs. Intex*, the court stated that any invention which has a technical contribution or has a technical effect and is not merely a computer program per se as alleged by the defendant and the same is patentable. Hence statement 3 is correct.

ENVIRONMENT AND ECOLOGY

85. With reference to India's Desert National Park, which of the following statements are correct?

1. It is spread over two districts.
2. There is no human habitation inside the Park.
3. It is one of the natural habitats of Great Indian Bustard.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1,2 and 3

Ans. (c) : Statement 1 is correct: The Desert National Park is situated in the Jaisalmer and Barmer districts of Indian state of Rajasthan stretched in 3162 km^2.
Statement 2 is correct: The park's vegetation is revered and protected by the local communities specially the Bishnois.
Statement 3 is correct: The endangered great Indian bustard is a magnificent bird found in relatively fair numbers.

86. Consider the following statements:

1. **36% of India's districts are classified as "overexploited" or "critical" by the Central Ground Water Authority (CGWA).**
2. **CGWA was formed under the Environment (Protection) Act.**
3. **India has the largest area under groundwater irrigation in the world.**

Which of the statement given above is/are correct?